

**CTCH 605 Project 4**  
**Introduction to Cybersecurity**

**Technical Controls Report**  
**Template**

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**Version 2.0**

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# 1. Executive Summary

FICBANK requested a technical security control plan after previous assessments identified exposed network services, critical vulnerabilities, weaknesses in centralized monitoring, and indicators of suspicious activity. Rather than performing another vulnerability assessment, the purpose of this project is to recommend practical controls that reduce risk while improving the organization's ability to detect and respond to cyber threats.

The recommendations in this report are based on the earlier network assessment, detective monitoring analysis, and incident investigation. Those activities showed exposed remote access services, patch management concerns, suspicious HTTP activity, and gaps in centralized logging. Together, these findings indicate that FICBANK would benefit from a layered security strategy instead of relying on single control.

The proposed controls include next-generation firewalls, network segmentation, multi-factor authentication, endpoint detection and response, vulnerability management, centralized logging through a SIEM, secure backup practices, and stronger identity management. Each recommendation includes an explanation of why the control is important and how it should be validated after implementation. Implementing these controls will help reduce the organization's attack surface, improve visibility across the environment, strengthen regulatory compliance efforts, and support long-term operational resilience.

## 2. Technical Control Scope

The scope of this implementation includes all major systems shown in the FICBANK network diagram, including perimeter security devices, virtual infrastructure, core servers, user endpoints, payment systems, communication systems, and backup infrastructure. The objective is to apply defense-in-depth by placing multiple complementary controls throughout the environment.

| System            | Business Function      | Expected Services                 |
|-------------------|------------------------|-----------------------------------|
| Firewall          | Protect network edge   | VPN, IDS/IPS, stateful inspection |
| Routers           | Route internal traffic | ACLs, routing                     |
| ESXi Host         | Virtualization         | HTTPS, SSH                        |
| Domain Controller | Authentication         | Kerberos, LDAP                    |
| DNS               | Name resolution        | DNS                               |

|                   |                    |                           |
|-------------------|--------------------|---------------------------|
| Exchange          | Corporate email    | SMTP, HTTPS               |
| Backup Server     | Recovery           | Backup software           |
| Office PCs        | User productivity  | Windows services          |
| POS               | Payment processing | Encrypted payment traffic |
| Field Workstation | Remote access      | VPN                       |

Although every system requires protection, the highest priority should be the Domain Controller, Exchange Server, backup environment, and Internet-facing devices because they present the greatest impact to business operations if compromised.

### 3. Technical Control Implementation Details

#### Perimeter Security

The perimeter firewalls should be upgraded to next-generation Firewall. Firewall rules should follow the principle of least privilege by allowing only approved business traffic. This recommendation addresses the exposed services discovered during the earlier assessment and reduces opportunities for unauthorized access. Implementing perimeter security provides FICBANK with a strong first layer of defense against external cyber threats

#### Network Segmentation

Critical servers, office workstations, backup systems, and point-of-sale devices should remain separated into dedicated VLANs. Access between VLANs should be controlled with ACLs so that users only communicate with resources required for their job functions. Segmentation limits lateral movement if a workstation becomes compromised. Implementing this control helps protect sensitive financial data, reduce the spread of malware and ransomware, and support the principle of least privilege.

#### Identity and Access Management

Administrative accounts should require multi-factor authentication and role-based access control. Password complexity, account lockout, and privileged account monitoring should be enforced through Active Directory. These controls protect against brute-force and password-spraying attacks, while audit logging enables security administrators to monitor authentication events and investigate suspicious activity.

## DNS and Exchange

DNS logging should be enabled and zone transfers restricted. Exchange should use anti-malware filtering, TLS encryption, and MFA for administrators. Email remains a common attack vector, making these protections essential. These controls reduce the likelihood of phishing attacks, credential theft, and unauthorized access to sensitive communications.

## Endpoint Protection

All workstations should use EDR, antivirus, disk encryption, USB device restrictions, and automated patching. Earlier assessments highlighted the importance of reducing vulnerabilities before attackers can exploit them. Implementing this control helps reduce the likelihood of successful attacks while improving visibility into endpoint activity.

## Backup and Recovery

Backups should be encrypted, isolated from the production network, and tested regularly. Offline or immutable copies provide additional protection against ransomware attacks. A well-designed backup and recovery strategy improves business continuity by allowing the organization to recover important systems and resume operations with minimal downtime.

## Vulnerability Management

Monthly authenticated Nessus scans should be performed across the environment. Findings should be prioritized using risk ratings, remediated according to defined timelines, and rescued to verify successful correction. By continuously monitoring systems and applying security updates in a timely manner, FICBANK can reduce its attack surface, improve system security, and strengthen its overall cybersecurity posture.

## Centralized Monitoring

A SIEM solution such as Microsoft Sentinel or Splunk should collect logs from servers, endpoints, firewalls, and network devices. Correlation rules and alerts should notify administrators of suspicious events in near real time. By implementing centralized monitoring, FICBANK can improve visibility in its environment, detect threats more quickly, and support more effective responses to the incident.

# 4. Test and Evaluation Plan

## Firewall Validation

Use Nmap to verify that only approved ports remain accessible and confirm firewall logs record blocked connection attempts.

## Vulnerability Management

Perform authenticated Nessus scans before and after remediation to verify that missing patches and configuration weaknesses have been corrected.

## Identity Controls

Test MFA with authorized and unauthorized users, review account lockout behavior, and verify RBAC permissions.

## Segmentation

Attempt communication between restricted VLANs to confirm ACLs prevent unauthorized access.

## SIEM

Generate failed logins, malware alerts, and administrative changes to verify successful collection, correlation, and alerting.

## Backup Recovery

Restore selected files and an entire system image to verify backup integrity and recovery objectives.

## Endpoint Security

Use the EICAR test file and review EDR alerts to confirm malware detection and response.

# 5. Conclusion

This project demonstrates how technical security controls can significantly strengthen FICBANK's cybersecurity posture. Building on the previous network assessment, monitoring analysis, and incident investigation, the recommended controls provide multiple layers of protection across the environment. Regular validation, continuous monitoring, vulnerability management, and user awareness will help ensure that the controls remain effective as threats evolve.

